

CricIntel-AI Project Report

1. Problem Statement

IPL franchises generate significant revenue through ticket sales, but pricing, marketing, and operational decisions are often based on experience rather than analytics. CricIntel-AI helps answer questions such as which matches deserve premium pricing, which venues and cities generate the highest revenue, which customer segments spend the most, when fans book tickets, which seat categories perform best, and how future ticket demand can be estimated. The project converts raw IPL ticket sales data into actionable business intelligence.

2. Why CricIntel-AI is Different

Unlike most IPL analytics projects that focus on player statistics, match prediction, batting, or bowling analysis, CricIntel-AI focuses on the business side of cricket. It analyzes ticket sales, customer segmentation, revenue optimization, booking behaviour, stadium occupancy, demand prediction, pricing intelligence, and business dashboards, making it highly relevant for Data Analyst and Business Intelligence roles.

3. Tech Stack & Datasets

Tech Stack:

- Python
- Pandas
- NumPy
- Matplotlib
- FuzzyWuzzy
- MySQL
- SQL
- Power BI
- Jupyter Notebook
- VS Code
- Git & GitHub

Datasets:

1. Matches
2. Customers
3. Ticket Sales

These relational datasets include match details, customer information, booking transactions, pricing, payment methods, loyalty tiers, venues, cities, and capacities.

4. Project Workflow

Collect synthetic IPL datasets.

Design relational tables in MySQL.

Perform SQL analytics, reporting queries, views, and validation.

Load cleaned data into Python with Pandas.

Clean and standardize the data.

Perform Univariate, Bivariate, and Multivariate EDA.

Engineer business features such as Lead Time and Match Demand Score.

Export processed datasets.

Build interactive Power BI dashboards.

Generate business insights and recommendations.

5. Data Flow

Raw CSV Files



MySQL Database



SQL Analysis & Validation



Cleaned Data Export



Python (Pandas)



Data Cleaning



Feature Engineering



Exploratory Data Analysis



Processed DataFrames



Power BI Dashboard



Business Decision Making

6. Key Business Insights

Revenue by Seat Category, Venue, City, Payment Method, Loyalty Tier, Age Group, Gender, Favorite Team, Occupation, Booking Behaviour, Occupancy, Demand Scoring, Pricing Outlier Detection, and Correlation Analysis.